

OSTEOACTIV® 3-in-1 Joint Formula Tablet Active Ingredients: Each tablet contains - Glucosamine HCl 250mg Methylsulfonylmethane (MSM) 250mg Chondroitin Sulfate Sodium 220mg (equivalent to 200mg of Chondroitin Sulfate) Presentations: Bottle of 100's. Boxes of 10's x 2, 10's x 4 and 10's x 12 in blister pack. Product Descriptions: White to off-white coloured, round and biconvex uncoated tablet. Pharmacodynamics: Glucosamine is a natural substance found in chitin, mucoproteins, and mucopolysaccharides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body; Glucosamine is also present in tendons and ligaments. Glucosamine must be synthesized by the body but the ability to do this declines with age. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis. Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity. Methylsulfonylmethane (MSM) is an organic sulfur-containing compound that occurs naturally in a variety of fruits, vegetables, grains and animals including humans. It blocks the inflammatory process and enhances the activity of cortisol, a natural anti-inflammatory hormone produced in the body. MSM is an effective natural analgesic and has been used as an application for pain relief over arthritis joints. Chondroitin is an acid mucopolysaccharide that is a constituent of most cartilaginous tissues. It is given orally in reactive arthritides such as gonococcal arthritis and given with Glucosamine for its supposed chondroprotective action in bone, joint, and connective tissue disorders. Chondroitin is an important structural component of cartilage and provides much of its resistance to compression. This combination therapy works synergistically to stimulate cartilage cell growth and provide strength to cartilage. Pharmacokinetics: Absorption: After oral administration, bioavailability of Glucosamine is low due to first-pass hepatic metabolism-26%. The gastrointestinal absorption is close to 90%. Distribution: Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins). Volume of distribution is 2.5 liters. Metabolism: Liver, extensive. The first-pass effect in the liver in which more than 70% of Glucosamine is metabolized. Excretion: Renal excretion 10%; Faeces 11%. Metabolism and excretion of MSM found the primary metabolite became detectable in serum approximately 2 hours after ingestion of MSM. With continued MSM ingestion, metabolite maintained a steady concentration in the serum. When MSM was stopped after 14 days, the mean metabolite concentration declined slowly over the subsequent 96 hours, and only trace amounts were detectable after 5 days. The decline in serum metabolite was linear, and its half-life appeared to be about 38 hours. Metabolite has been shown to persist in the blood up to 5 times longer than MSM. Earlier studies using high molecular weight Chondroitin concluded that there was no significant absorption of the high molecular weight version of Chondroitin. More recent studies demonstrate that there is probably significant absorption of low molecular weight Chondroitin. Absorption appears to occur from the stomach and small intestine. There is also an indication that some Chondroitin, after absorption, does enter the joint space. In a study using depolymerized shark Chondroitin, approximately 30% of the administered drug (as high and low molecular weight derivatives) was absorbed. Once absorbed, the Chondroitin is incorporated into glycosaminoglycan-rich tissues such as joint cartilage. Indications: As adjuvant therapy for osteoarthritis. Dosage and Administration: For oral administration only. Adults: Light or moderate osteoarthritis symptoms: Take 2 tablets 2 times daily before food for a period of at least 6 weeks (or as directed by healthcare professional). Severe osteoarthritis symptoms: Take 2 tablets 3 times daily before food for a period of at least 8 to 12 weeks (or as directed by healthcare professional). Maintenance: Take 2 tablets twice daily before food. The maintenance therapy should be followed for 3 to 4 months (or as directed by healthcare professional). The treatment of osteoarthritis should be repeated every other	
6 months or less (or as directed by healthcare professional). Children: Safety and effectiveness have not been established in children. Symptoms and Treatments of Overdose: No cases of accidental or intentional overdose are known. The animal acute and chronic toxicological studies indicate that toxic effects and symptoms of toxicity are not likely to occur, even after high overdose. Contraindications: Hypersensitivity to Glucosamine, MSM or Chondroitin. As Glucosamine is obtained from seafood (shellfish), the product should not be given to patients who are allergic to shellfish. Warnings and Precautions: Glucosamine treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2 to 3 weeks. Therefore, it is advisable to take an analgesic or anti-inflammatory drug if required during the first 2 to 3 weeks of therapy with Glucosamine. Administration during the first three months of pregnancy must be avoided. Safety and effectiveness have not been established in children, therefore children should avoid using Glucosamine. The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision. A doctor should be consulted in order to exclude the presence of other joint conditions/diseases for which an alternative treatment should be considered. This product contains Glucosamine derived from seafood, therefore should not be given to patients who are allergic to shellfish. This product contains Chondroitin from bovine source. No effects on the ability to drive or to operate machines are expected. Use in Pregnancy and Lactation: Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mothers if the potential benefit to the mother justifies the potential risk to the fetus. Administration during the first 3 months of pregnancy must be avoided. Interactions with Other Medicaments: Effects on glucose metabolism and antidiabetic agents: It has been hypothesized that Glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of peripheral glucose uptake. Glucosamine may increase insulin resistance and consequently affect glucose tolerance. It may reduce antidiabetic agent effectiveness eg when used with these antidiabetic agent: Acarbose, Acetohexamide, Chlorpropamide, Glipizide, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride, Tolbutamide and Troglitazone. Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely or more frequently. Reduced effectiveness when used with Glucosamine: Doxorubicin, Etoposide and Teniposide. Warfarin: Elevations of International Normalized Ratio (INR) serum values and potentiation of anticoagulant effects. If concomitant therapy is necessary, the patient's INR should be more closely monitored. Undesirable Effects: Cardiovascular: Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established. Central nervous system: Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%). Gastrointestinal: Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with Glucosamine. Skin: Skin reactions such as erythema and pruritus have been reported with therapeutic administration of Glucosamine. Storage: Store below 30°C. Protect from light, heat and moisture. Keep container tightly closed. Keep out of reach of children. Jauhkan daripada capaian kanak-kanak. Shelf Life: 3 years from the date of manufacture. Manufacturer and Product Registration Holder WINWA MEDICAL SDN. BHD. 198101009677(75794-A) 1952, Lorong I.K.S. Bukit Minyak 3, Taman I.K.S. Bukit Minyak, 14000 Bukit Mertajam, Pulau Pinang, Malaysia. MAL12045022X BRU16042031NP Date of Revision: 30 December 2025 X.TOT-I.07	

Size: 75mm x 165mm



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